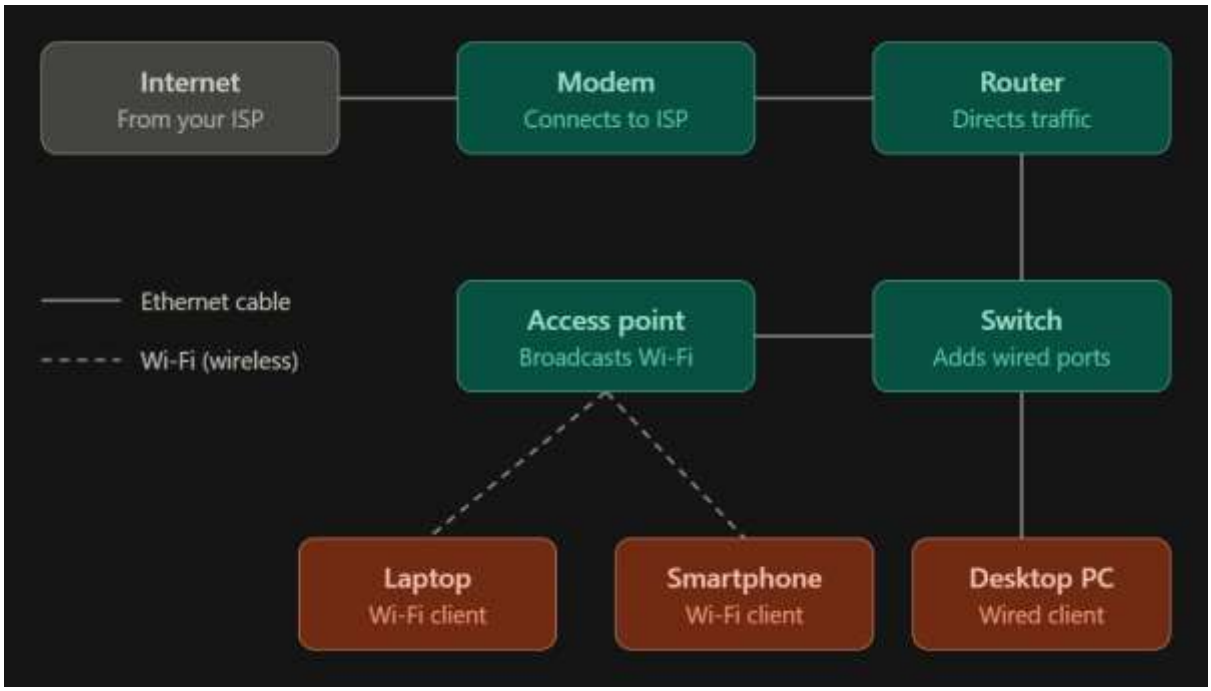
Day 1: Introduction to Networking Hardware Theory (1 hour): Overview of modems, routers, switches, access points, and network cables.

1.Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

ANS:

Here's a typical home Wi-Fi setup, from the internet connection down to the devices.



The diagram follows the path your data takes:

- Modem: converts the signal from your internet provider into something your home network can use.
- Router: directs traffic between your devices and the internet, and hands out IP addresses.
- Switch: adds extra wired ports, so more devices can plug in by Ethernet cable.
- Access point: broadcasts the Wi-Fi signal so wireless devices like the laptop and smartphone can join.

2.List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it.

ANS:

Device	Main function	Where you'd see it
Modem	Converts the signal from your internet provider (cable, DSL, or fiber) into a form your network can use	The box your ISP installs, connected to the wall outlet or fiber line at home
Router	Directs traffic between your devices and the internet, and assigns each device an IP address	At home, sharing one internet connection across phones, laptops, and TVs
Switch	Adds wired ports so many devices can share one network and talk to each other	In an office, connecting desktops, printers, and IP cameras to one network
Access point	Broadcasts a Wi-Fi signal for wireless devices to join a wired network	Ceiling-mounted in offices, hotels, or large homes to extend Wi-Fi coverage
Network cable	Carries data between devices as electrical or light signals	The Ethernet cable running from your router to a desktop PC or gaming console

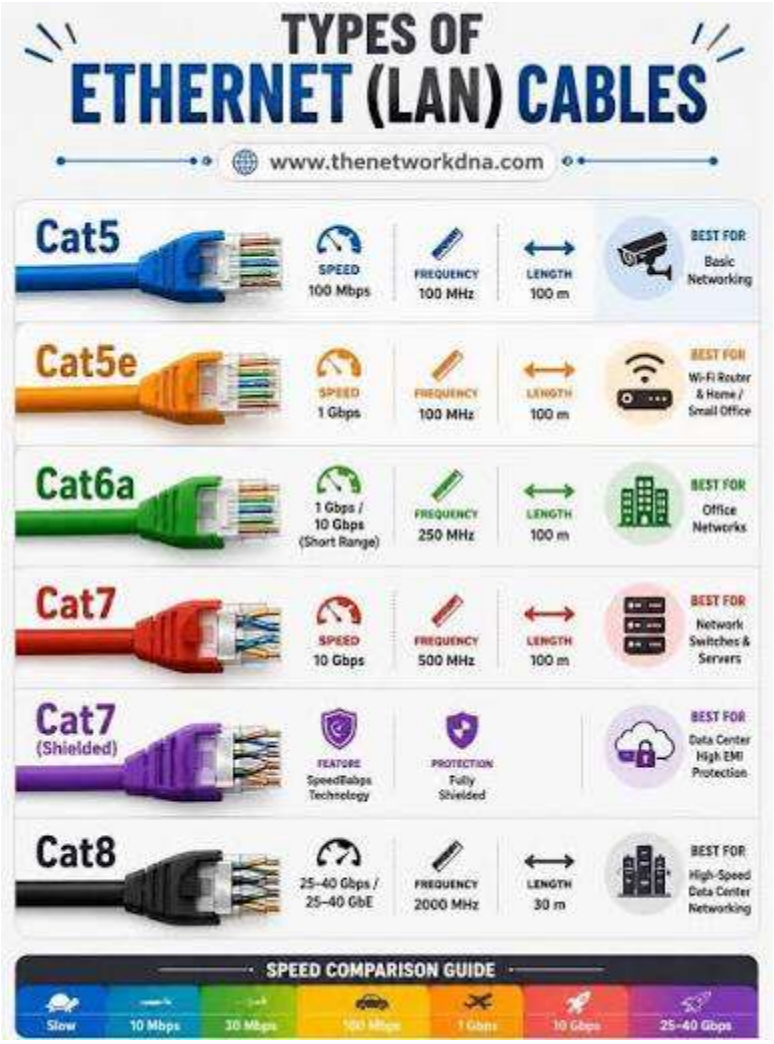
3.Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

ANS:

1.Ethernet cable:

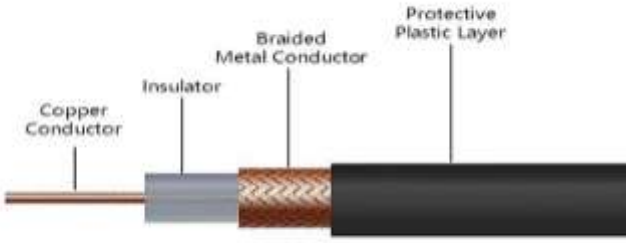
Description: Ethernet cables are commonly used to connect computers, routers, and switches in homes, schools, and offices to form a local area network (LAN).

These are the different versions of ethernet cable as shown in the picture below.



2. Coaxial Cable:

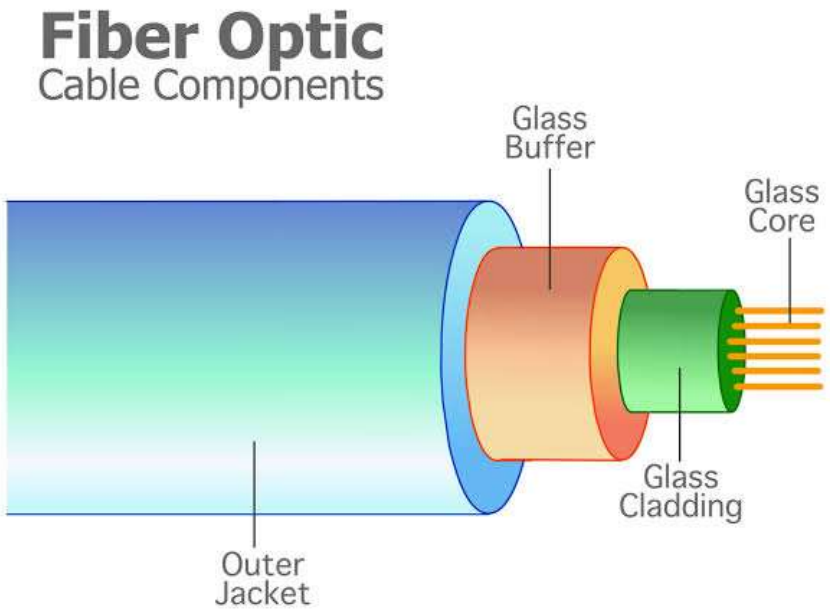
Description: Coaxial cables are commonly used for cable television, satellite connections, and cable broadband internet**.**



3.Fibre-Optic Cable:

Description: Fibre-optic cables transmit data using pulses of light and are commonly used for high-speed internet, telecommunications, and long-distance data transmission**.**

Here is a photo of a fibre optic cable with its parts or insides with their names.



4.Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle:

ANS:

Feature	Router	Switch
Main use	Connects different networks, such as your home network and the internet	Connects devices within one local network
Typical location	At the network edge, right after the modem	Inside the network, between the router and wired devices
How it handles data	Reads IP addresses and chooses the best path for each packet	Reads MAC addresses and sends frames only to the port of the intended device
Extra services	Usually provides DHCP, NAT, and a firewall	Usually none, though managed switches add VLANs and monitoring

Feature	Router	Switch
Network layer	Layer 3 (network)	Layer 2 (data link)
Wi-Fi	Home routers often include built-in Wi-Fi	Wired ports only